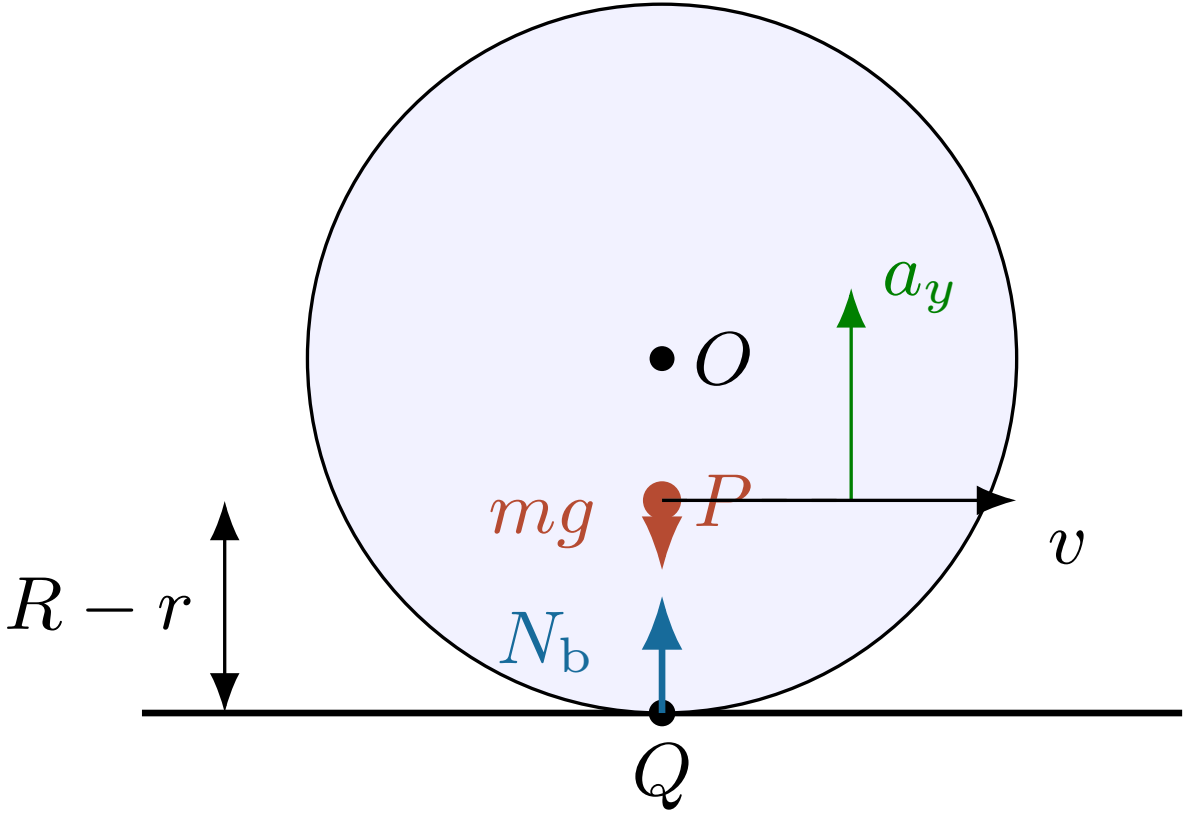

$$\theta = \pi, \quad f = 0$$